

Gregory Lim
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Education

Singapore University of Technology and Design
Bachelor of Engineering in Computer Science

Singapore
Sep 2022 – May 2026

Experience

Open Source Contributor
Google DeepMind

Jul 2026 – Jul 2026
Singapore

- Solved 19-year-old open mathematics problem by proving finite, simple, K_3 -free, induced- P_5 -free graphs are well-totally-dominated, and submitting machine-verified Lean proof in Formal Conjectures under PR 4455.

AI Engineer Intern
Creatz3D

May 2025 – Sep 2025
Singapore

- Built industry-first geometry-constrained mesh thickening models.

Proficiencies

- **Languages.** C, CSS, HTML, JS, Python, Shell, TeX, and more.
- **Tools.** Git, Gradio, HF, LangChain, n8n, NumPy, pandas, PyTorch and more.

Projects

Quantization-Envelope Rejection Sampling

WIP

- Invented Las Vegas LM decoding algorithm that enables faster inference.

Vibe Decoding

github.com/glyaea/vibe-decoding

- Invented single-hyperparameter, $O(Vd)$ -time LM decoding rule that enables more creative inference.
- Outperformed macro F1 and MRR of Qwen 2.5 0.5B, Gemma 2 2B, and Llama 3.1 8B on greedy decoding by up to 10.03% on BoolQ, and up to 5.12% on NumerSense.

SUTDoko

sutdoko.vercel.app

- Built typo-resistant SUTD room look-up using decoding-constrained transformer-style SLM.
- Outperformed accuracy of official SUTD Assistive Maps Telegram bot by 100% on subset of test set.
- Made inference smoother and over 10 times faster by making it cached, debounced, and incremental.

Recognition

- Information Systems Prize, Singapore University of Technology and Design & Singtel, 2024.
- Mathematics and Science Book Prize, Temasek Secondary School, 2017.